PHARMACOECONOMICS





Lecture 2

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Introduction

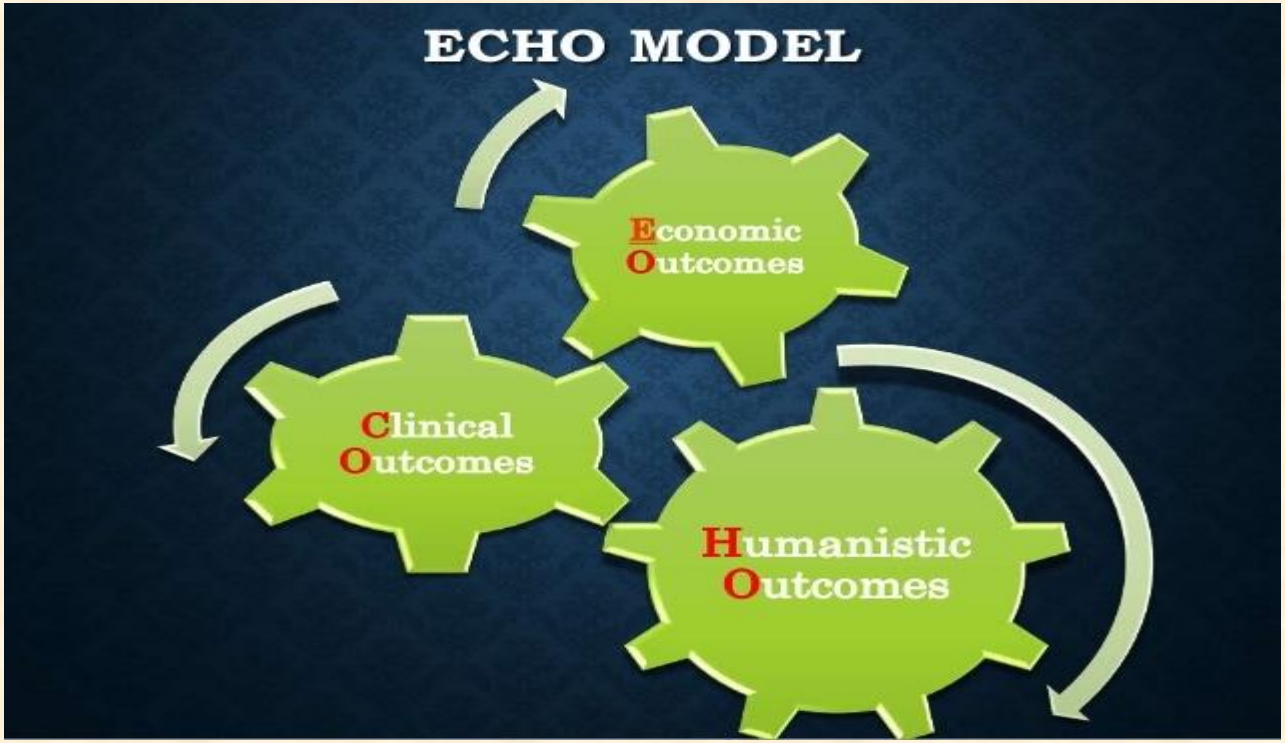
- Health economics is the science of assessing cost and benefits (outcomes) associated with the use of healthcare.
- Pharmacoeconomics is a branch of health economics which compares the value of one drug or a drug therapy to another.
- Pharmacoeconomics is defined as the application of economic analysis to the use of pharmaceutical products, services and programs, which frequently focuses on the costs (inputs) and consequences (outcomes) of that use.
- Pharmacoeconomics is defined as the process of identifying, measuring, and comparing the costs, risks, and benefits of programs, services, or drug therapies and determining which alternative produces the best health outcome per money spent.

- Pharmacoeconomics: is a set of methods to evaluate the

1. Economic,
2. Clinical, and
3. Humanistic

Outcomes (ECHO) of pharmaceutical products and services (or any health care service)

➢ This is called ECHO model.



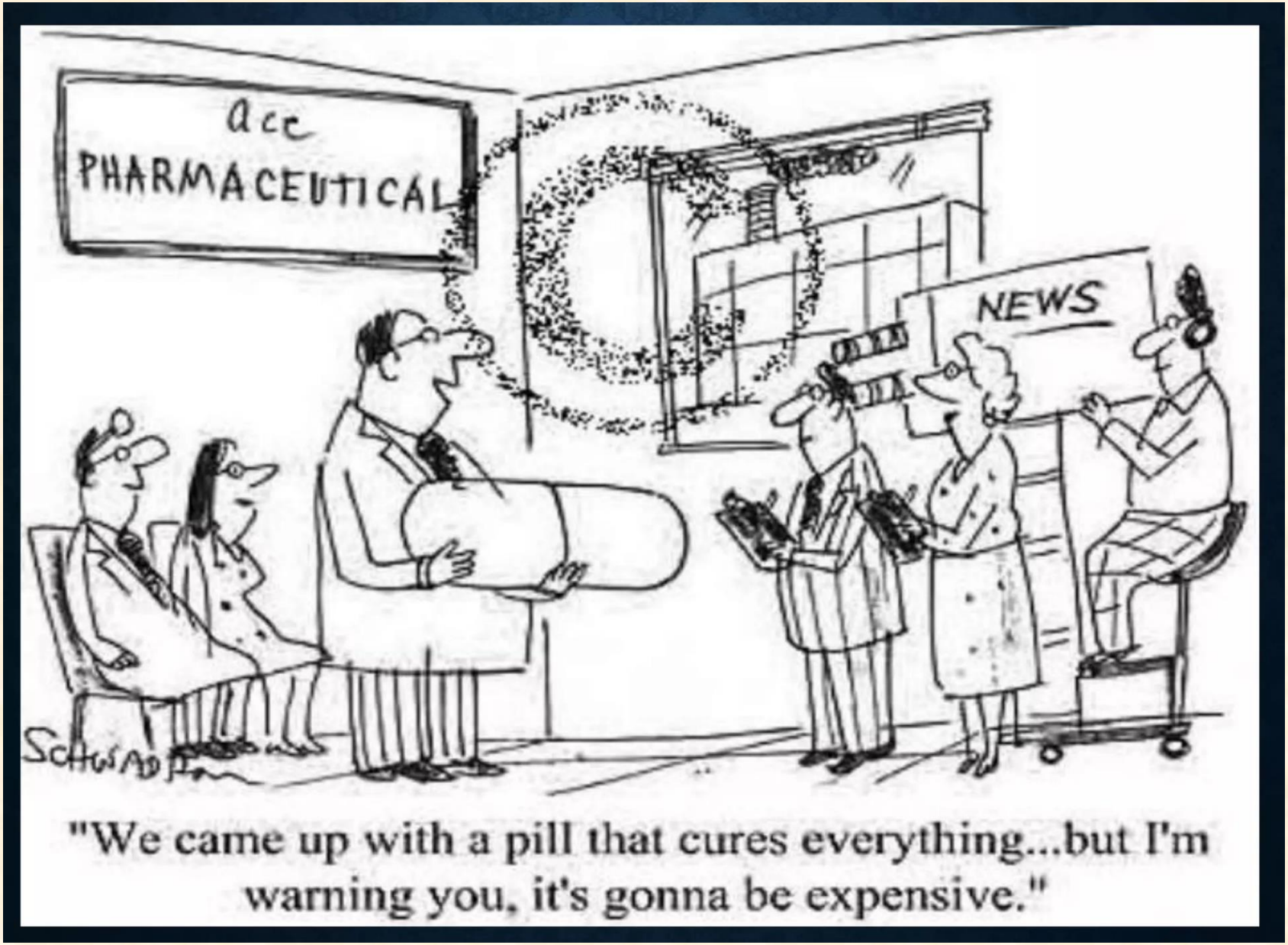
- Pharmacoeconomic evaluation is a comparative analysis of alternative healthcare interventions in terms of both their costs and outcomes.

- The primary objective of pharmacoeconomics is to maximize the net health benefit derived from the use of finite (limited) healthcare resources.

- – costs: resources consumed
- – outcomes: clinical, economic, and humanistic
- – interventions: pharmaceutical products, pharmaceutical services, non-drug therapies, public health programs, etc

WHY STUDY PHARMACOECONOMICS ??

- Helps to decide which drug to develop
- To estimate and understand the full impact of new therapy
- To make an informed decision regarding appropriate use of drug which have been developed
- **To make the best use of limited resources**



"We came up with a pill that cures everything...but I'm warning you, it's gonna be expensive."

GOALS:



- ✓ To determine which healthcare alternatives provide the best healthcare outcome in terms of money spent

- ✓ To improve the allocation of resources for pharmaceutical products and services

Pharmacoeconomic Analysis

Steps for conducting a pharmacoeconomic analysis involve:

1) Define the problem and state the objective

2) Identify the perspective and alternative interventions to be compared

3) Identify, measure and value costs of each alternative

4) Identify and measure outcomes of each alternative

5) Use **discounting** and **sensitivity** analysis when appropriate

1. Define the problem and state the objective

- Identify the disease state and what aspect you want to deal with.

- i.e. What is the most cost effective method for controlling glucose in the treatment of type II diabetes?

2. Identify the perspective

- That is, who will be utilizing (i.e. use) the information to make what decisions.
- This will guide you in choosing the relevant costs and benefits.

Perspectives of Evaluation

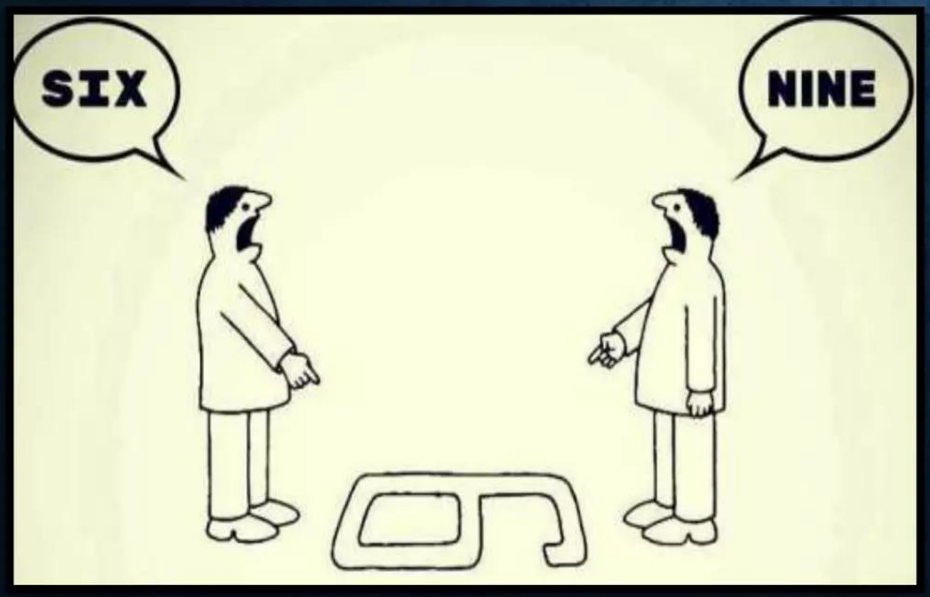
Common perspectives include:

- Patient perspective – Portion of cost not covered by Insurance (Out-of-Pocket costs, lost income, transportation)
- Provider perspective – eg. Hospitals (hospital stay costs, treatment of adverse events & complications, pharmacy, supplies)
- Payer perspective – eg. Insurance companies, employers, or the government (hospitalization, pharmacy, nursing home care)
- Society perspective – All possible costs (all direct and indirect cost) including lost productivity.

PERSPECTIVES OF PHARMACOECONOMICS

- Patient perspective
- Provider perspective
- Payer perspective
- Societal perspective

After selection of perspective next step cost related measurements



1. Direct medical costs
2. Direct non medical costs
3. Indirect nonmedical costs
4. Intangible costs
5. Opportunity costs

3. Identify, Measure and Value costs

The value of the resources consumed by a program or drug therapy, is defined as **Cost**.

Healthcare costs are categorized as follows:

- Direct Medical Costs: Drugs, medical supplies, and equipment, laboratory and diagnostic tests, hospitalizations, and physician visits.
- Direct Nonmedical Costs: Transportation to and from healthcare facilities, extra trips to the emergency department, child or family care expenses, special diets, and various other out-of-pocket expenses.
- Indirect Nonmedical Costs: **Morbidity cost** (Loss of productivity, missed work due to illness), **Mortality cost** (Loss of years of service due to premature death).

- **Direct Medical Costs**: Costs of medical service

These include:

- **Fixed costs** or costs that do not vary immediately with the number of patients treated. E.g. capital costs of hospital building or equipment etc.

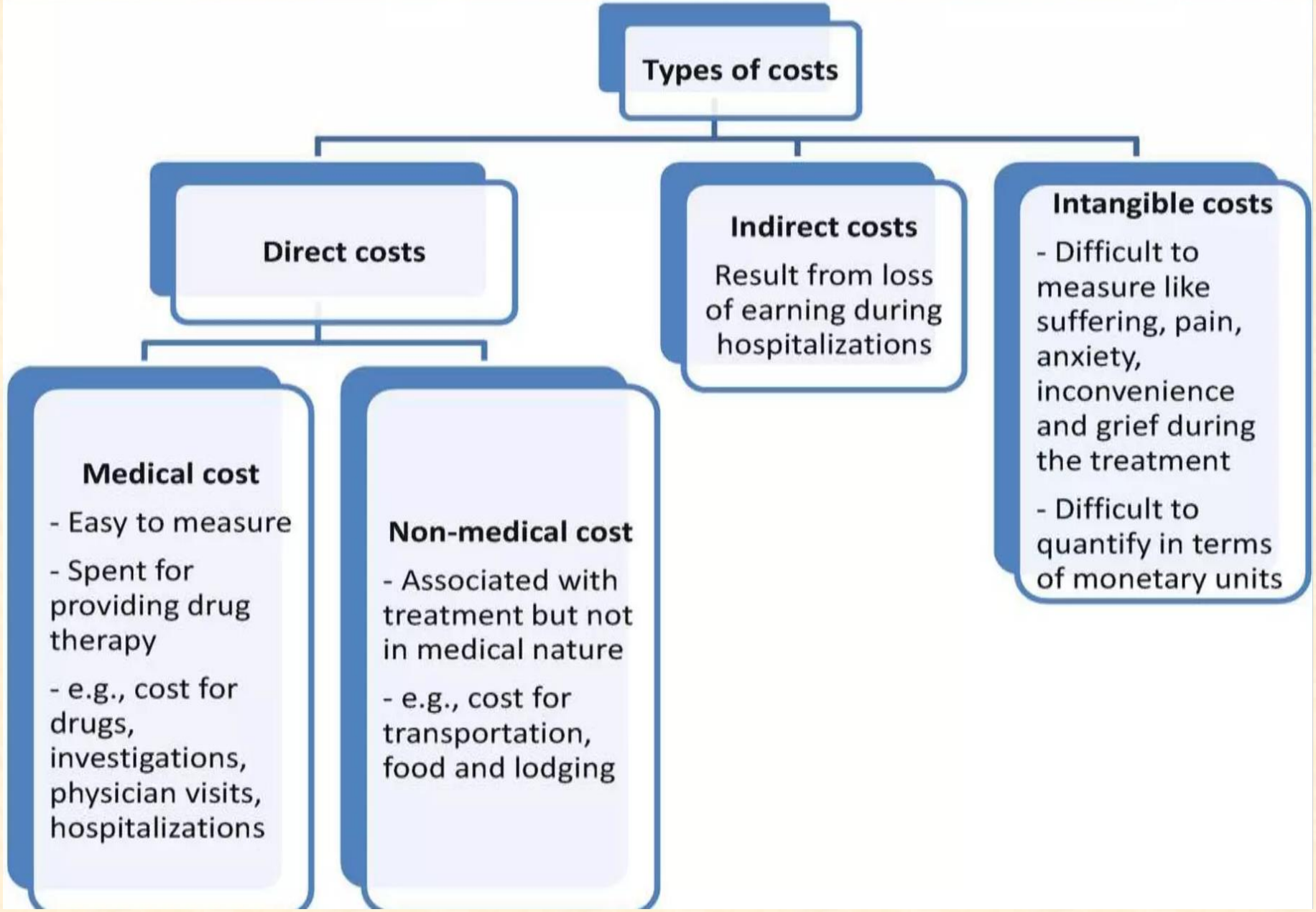
- **Variable costs** or costs that vary immediately with number of patients treated. E.g. costs of drugs, syringes, needles etc.

- **Direct non-medical costs**:
  - Costs incurred by the patient in receiving medical care. E.g. transportation to and from hospital.

- Intangible Costs: Nonfinancial outcomes of disease and medical care such as pain, suffering, inconvenience, and grief that occur because of an illness or the treatment of an illness.

- Opportunity Costs: Value (economic benefit) of the alternative therapy that was forgone (e.g. is the value of what you lose when selecting one therapy substitute over the subsequently best alternative.)
- Or (The amount lost by not using economic resources in its best alternative use (labour, capital, building, management etc.) • Resources invested in one area will be at expense of loss of another opportunity.

- Incremental Costs: The extra costs required to purchase an additional unit of effect.


Types of costs
Direct costs
Indirect costs
Result from loss of earning during hospitalizations
Intangible costs
- Difficult to measure like suffering, pain, anxiety, inconvenience and grief during the treatment
- Difficult to quantify in terms of monetary units
Medical cost
- Easy to measure
- Spent for providing drug therapy
- e.g., cost for drugs, investigations, physician visits, hospitalizations
Non-medical cost
- Associated with treatment but not in medical nature
- e.g., cost for transportation, food and lodging

The Idea of Opportunity Cost:

- In order to understand opportunity cost, you must realize that the opportunities that are lost because of limited resources are known as opportunity cost.

- Since resources are limited, every time you make a choice about how to use them, you are also choosing to forego other options.

- Economists use the term **opportunity cost** to indicate what must be given up to obtain something that's desired.

- A fundamental principle of economics is that **every choice has an opportunity cost**. If you spend your income on video games, you cannot spend it on movies. If you choose to marry one person, you give up the opportunity to marry anyone else.

- In short, opportunity cost is all around us.

Opportunity Cost Formula and Calculation

$$\text{Opportunity Cost} = FO - CO$$

where:

FO=Return on best foregone (given up) option
CO=Return on chosen

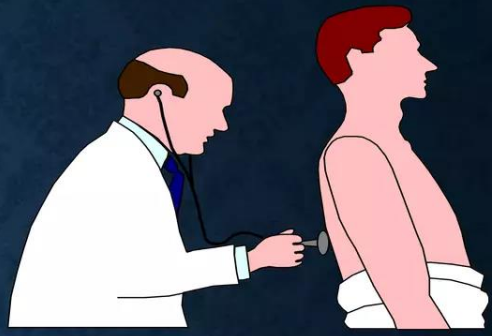
Example:

Assume the expected return on investment in the stock market is 12% over the next year, and your company expects the equipment update to generate a 10% return over the same period. The opportunity cost of choosing the equipment update over the stock market is (12%-10%=2%), which equals two percentage points. In other words, by investing in the business, you would forgo the opportunity to earn a higher return.

Direct Costs = Direct Medical Costs + Direct non-medical costs
Indirect Costs = Morbidity costs + Mortality costs
Total costs = Direct costs + Indirect costs + Intangible costs

	Definition	Example
Direct Medical	Cost for specific health care services	· Medications for COVID-19 · COVID-19 testing · COVID-19 vaccine · ED bill
Direct Non-Medical	Costs necessary to receive health care services	· Transport to the hospital
Indirect	Measures the lost of productivity	· Sick days from work · Fatigue from COVID making future work not as productive · Quarantine limiting the amount of work that can be done
Intangible	Reflects level of pain and suffering	· Anxiety from spreading COVID / condition getting worse · Flu like symptoms · Loss of taste / smell

PATIENT PERSPECTIVE



- All the relevant cost and consequences experienced by the patient

- Included costs:

  Direct

  Indirect

  Intangible

PROVIDER PERSPECTIVE



- Concerned with the expenses of providing products or services

- Included costs:

  -Direct costs only

PAYER PERSPECTIVE



- Social Security/Government, third party payers

  eg. private insurance companies and employers

- Included costs:

  -Direct costs

  -Indirect costs

  relevant to employers

  lost workdays

  lost productivity at work

SOCIETAL PERSPECTIVE



- The broadest of all perspectives that comprehensively evaluates all costs and consequences

- Considers the benefits to society as a whole

  Included costs:

  - Direct; overall cost of providing care
  - Indirect; loss of productivity

Measuring Costs

- Costs are measured over a relevant time period such as a month or year.

- The length of time used depends on the typical span of the illness. Acute diseases such as the flu would have a short span; while chronic or long-term illness such as depression or heart disease would span years.

Valuing Costs

- Determining the amount (dollar value) for each item (cost/benefit) listed can be difficult, especially for indirect and intangible costs.

- Sources of cost data include claims data, published price lists…….depends on your perspective.

4. Identify and measure outcomes of each intervention

Typical outcomes used include:

- cure rate (percent cured of illness)
- improved quality of life
- decreased incidence of morbidity
- years of extended life
- relief or reduction in symptoms
- no effect
- Adverse events (drug interactions and side-effects)
- mortality

Types of Outcomes

- **Intermediate** outcomes such as controlling sugar levels, blood pressure and cholesterol levels are indicators that a disease or event (like a stroke) is less likely to occur.

- **Final outcomes** would be measured as the reduction in the disease or events.

- Values can come from RCTs, literature, surveys or other data sources.

Another classification of the studied outcomes:

Clinical outcomes:

- Are the biologic or physical manifestations resulted from treatment
- are the primary measure of effectiveness for acute conditions and curative treatments
- measured in natural units (traditional clinical trial end points)
- these outcomes consider clinical data alone
- Examples: – number of lives saved/ increased life expectancy – number of deaths avoided/ decreased mortality – number of cases cured – laboratory values (such as reductions in hemoglobin A1c, reduction in mm HG, reductions in LDL-cholesterol levels etc.)

Humanistic outcomes:

- Are the consequences of treatment as perceived and reported by the patient

- Are effects of treatment on patient's health related quality of life (HRQOL)

- HRQOL is a combination of a person's **physical**, **mental** and **social** wellbeing; not just the absence of disease

- Clinical outcomes are **clinician**-assessed end points and humanistic outcomes are **patient**-reported functioning and well-being

• HRQOL has several **domains** (e.g. areas or fields)

• commonly measured domains include:
– physical health and functioning
– mental/emotional health and functioning
– social and role functioning
– pain and other symptoms such as nausea/vomiting

• Humanistic outcomes are the primary measure of effectiveness for chronic conditions and palliative treatments (i.e., reducing symptoms but not curing the underlying disease)

• Humanistic outcomes are most commonly used as end points for asthma, cancer, schizophrenia, arthritis, HIV/AIDS, depression, anti inflammatory, central nervous system, respiratory, and allergic conjunctivitis therapy areas.

Economic outcomes:

- Include reduction in resource use or savings (cost avoidance) resulted from medical intervention.

- Examples: – savings due to preventive strategy – increased productivity due to treatment.

Outcome Measures

Disease	*Indicator*	*Clinical Outcome*	*Humanistic Outcome*	*Economic Outcome*
Hypertension	BP	Renal failure Stroke MI Death	QOL	Cost/↓ mmHg BP Cost/stroke avoided Cost/life year saved
Hyperlipidemia	LDL levels	Angina MI Death	QOL	Cost/MI avoided Cost/point ↓ in LDL
Diabetes	A1C BG levels	Retinopathy nephropathy Death	QOL	Cost/change in A1C Cost/kidney transplant avoided
Asthma	FEV, peak flow	Exacerbation event Death	QOL	Cost/symptom free day

Some outcome definitions:

- Positive outcomes: Desired effect of a drug.
- Negative outcomes: treatment failure, ADRs, drug toxicity, or even death.
- Intermediate outcome: Can serve as a proxy for more relevant final outcomes.
- Final outcome: To get final outcome of reduced MI rate (lipid lowering agents are being used to decrease LDL levels which is an intermediate outcome).
- Pharmacoeconomic evaluations should include assessments of both **positive** and **negative** outcomes.

5. Use sensitivity and discounting analysis when appropriate

Sensitivity analysis

- When estimating costs and outcomes, you typically have a range of possible values.

- Sensitivity analysis requires that the results be recalculated at the different values to see if the conclusions change (e.g. Choose option A or option B)

Discounting analysis

- If cost are estimated based on dollars spent or saved in future years, another type of adjustment, **called discounting**, is needed. There is a time value associated with money.

- Most people (and businesses) prefer to receive money today, rather than at a later time. Therefore, **a dollar received today is worth more than a dollar received next year.**

- Discount rate, a term from finance, approximates the cost of capital by taking into account the **projected inflation rate** and estimates the time value of money.

Discounting

If the analysis spans more than a year, then the dollar values must be adjusted to a common point in time.

- Discounting adjusts future costs or benefits using an expected interest or ***discount rate.***
- Present Value = $\frac{\text{Future value}}{(1+r)^n}$

where r = discount rate (typically ranges from .03 to .06) and n = the number of years in the future.

Discounting example

- You wish to implement a diabetes DSM program which will cost you $1500 per year.
- The benefits from this program won't be evident for 2 years, so you want to evaluate it after 4 years.
- Use r = .05

- The most common discount rate currently seen in the literature is 5%, the approximate cost of borrowing money today.

Discounting Example

Year	Costs	PV
1	1,500	1,429
2	1,500	1,361
3	1,500	1,296
4	1,500	1,234
total	$6,000	$5,320


THANK
YOU